

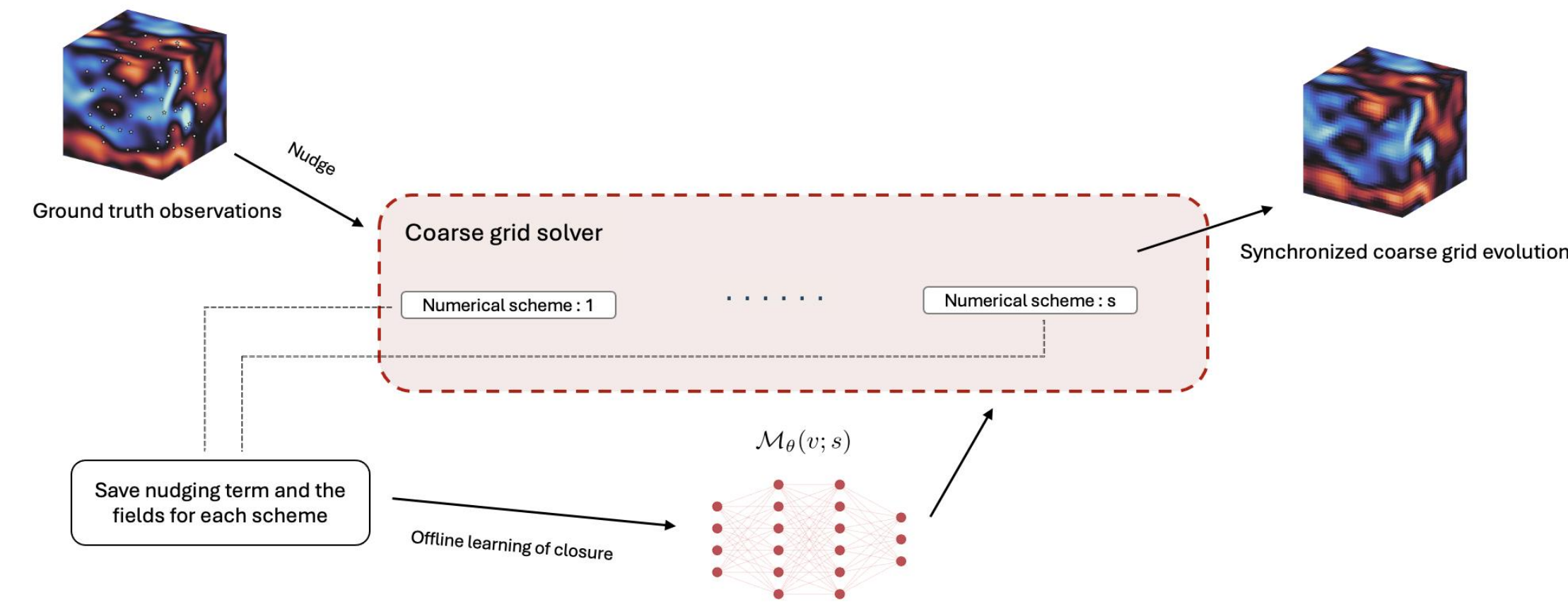
Nudging based turbulence closure modeling

Goal

This work develops a data-driven turbulence closure modeling framework for LES inspired by continuous data assimilation (nudging). Instead of training the closure *a-posteriori* by embedding a neural network inside the solver, we learn the nudging correction *a-priori* from coarse-grid LES synchronized to DNS observations.

The goal is to recover the forcing needed to match ground-truth turbulence statistics while maintaining long-term stability, without adjoints or backpropagation through the solver.

Methodology



Ground truth : $\partial_t u = \mathcal{N}[u]$

LES + Nudging :

$$\partial_t v = \mathcal{N}[v] + f_{smag} + \mu f_h(u_{obs} - v) \quad \text{"nudge"}$$

Learned closure :

$$\mathcal{M}_\theta(v; s) \approx (u_{obs} - v)$$

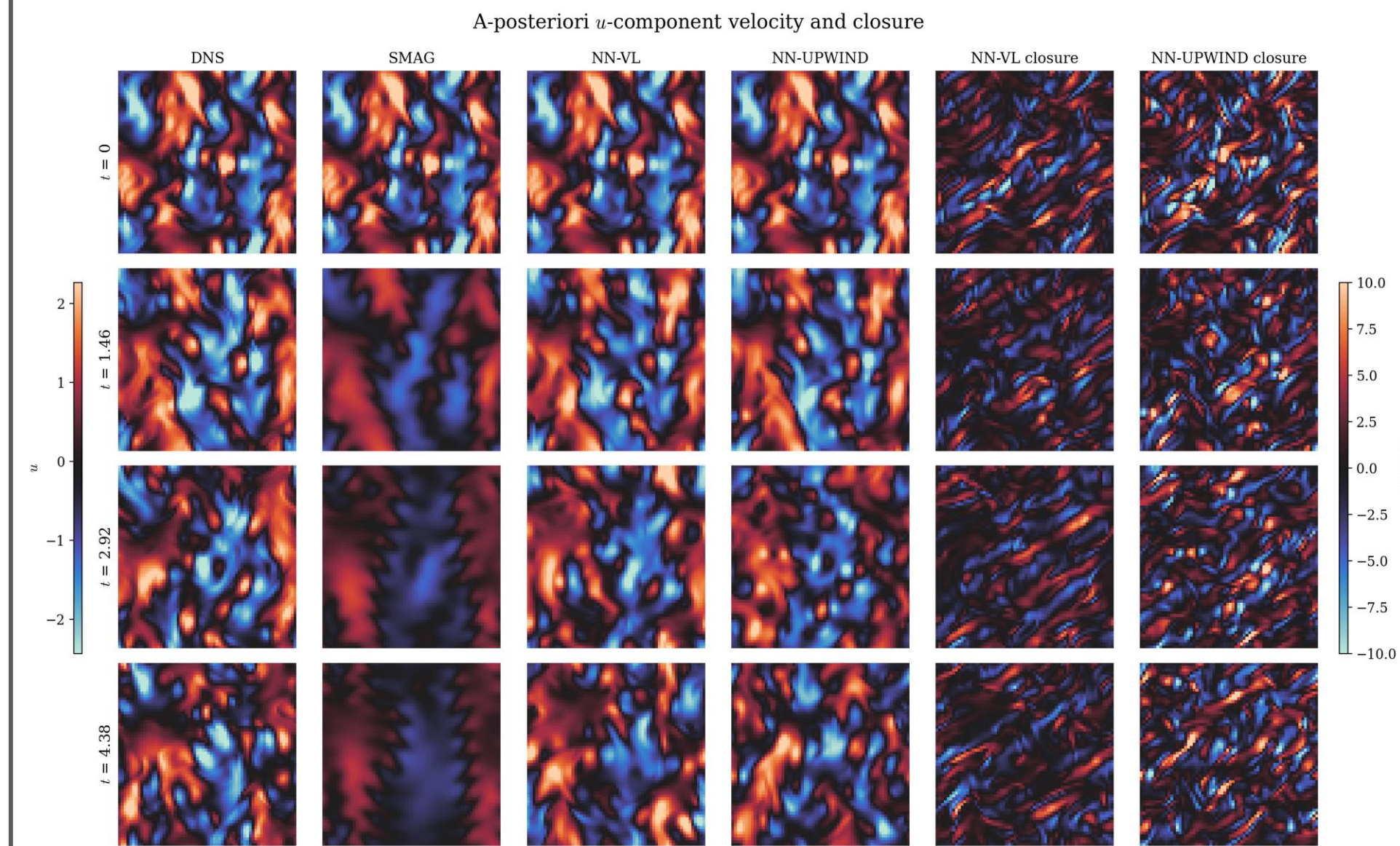
- s = encoding for different convective term schemes
- v = coarse grid field, u = ground truth / DNS

Observation : available at all CG points, no noise

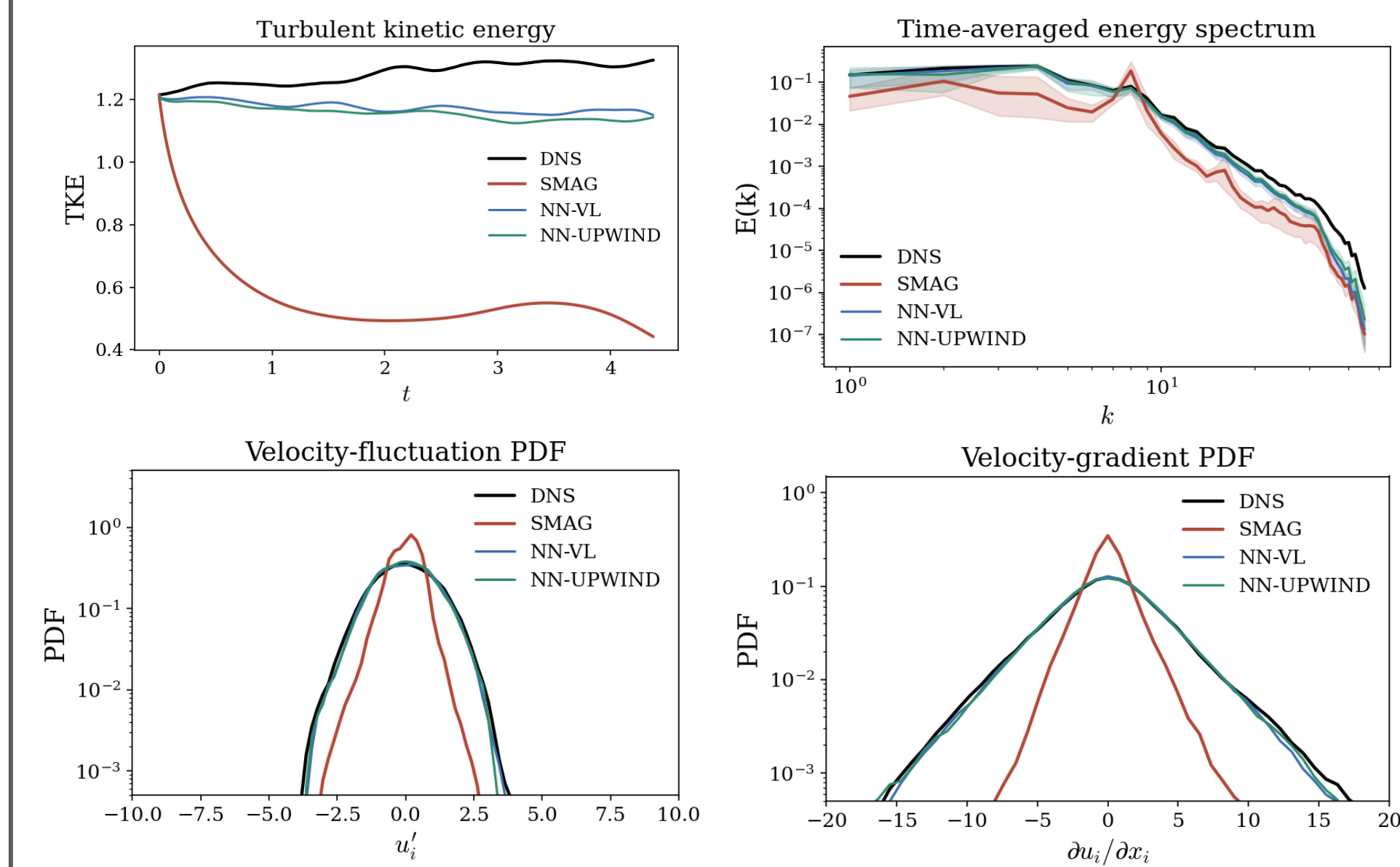
2D : Upwind and Vanleer (VL) and 3D : Central difference 2, 4, 6 (C2,4,6)
Only convective term is changed

Model : Swin Transformer + FiLM embeddings to encode different numerical schemes

Results : 2D HIT

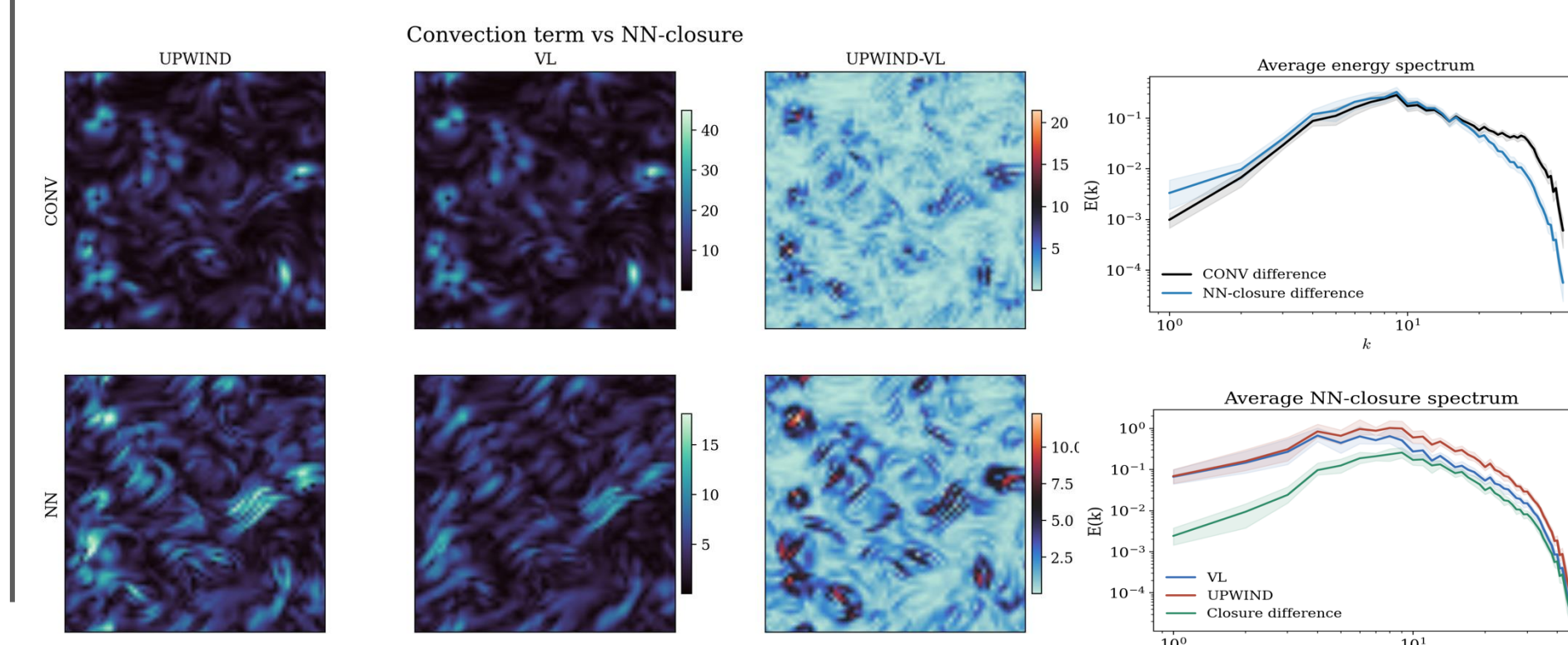


The model maintains the turbulence statistics better than traditional closure

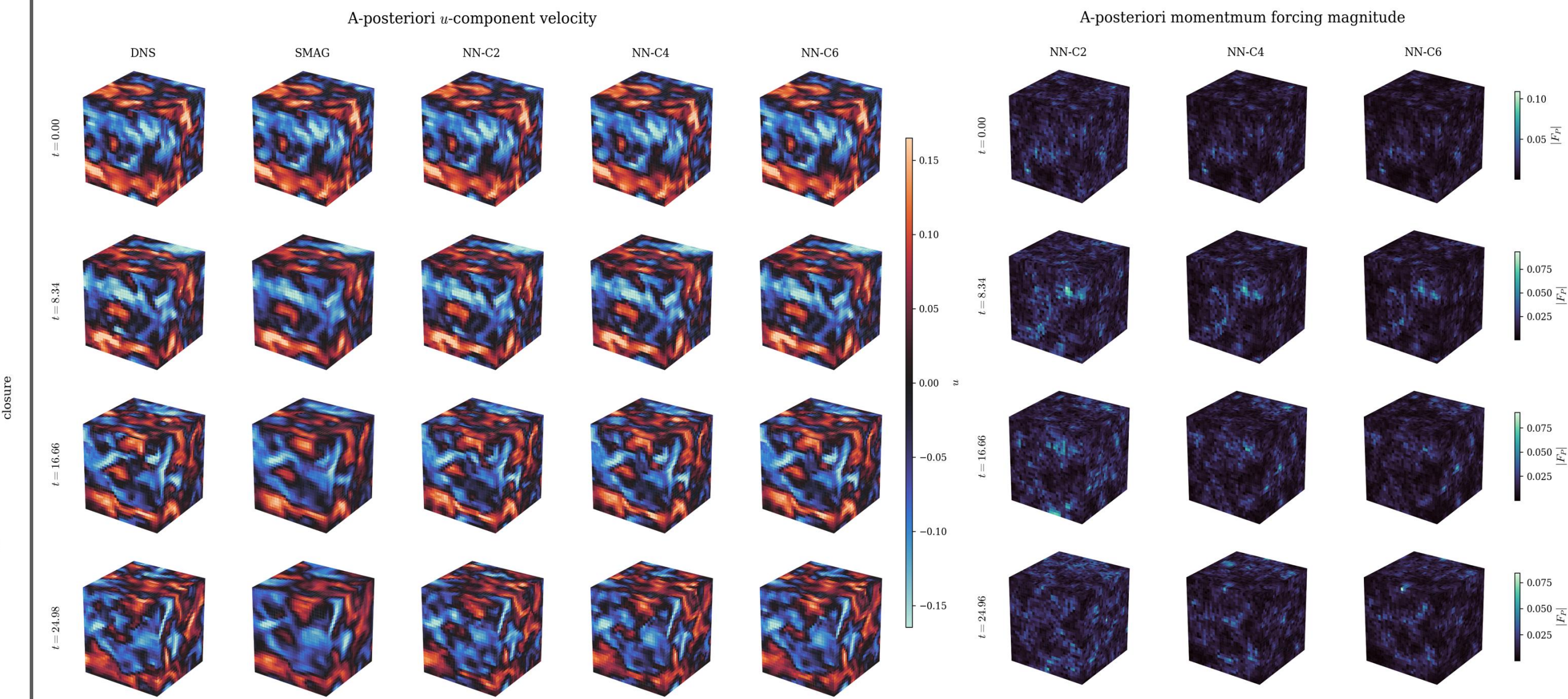


Convective term : $\mathcal{C}(v) = -(v \cdot \nabla)v$

$$\mathcal{C}^{s_1}(v) - \mathcal{C}^{s_2}(v) \approx \mu(\mathcal{M}_\theta(v, s_1) - \mathcal{M}_\theta(v, s_2))$$

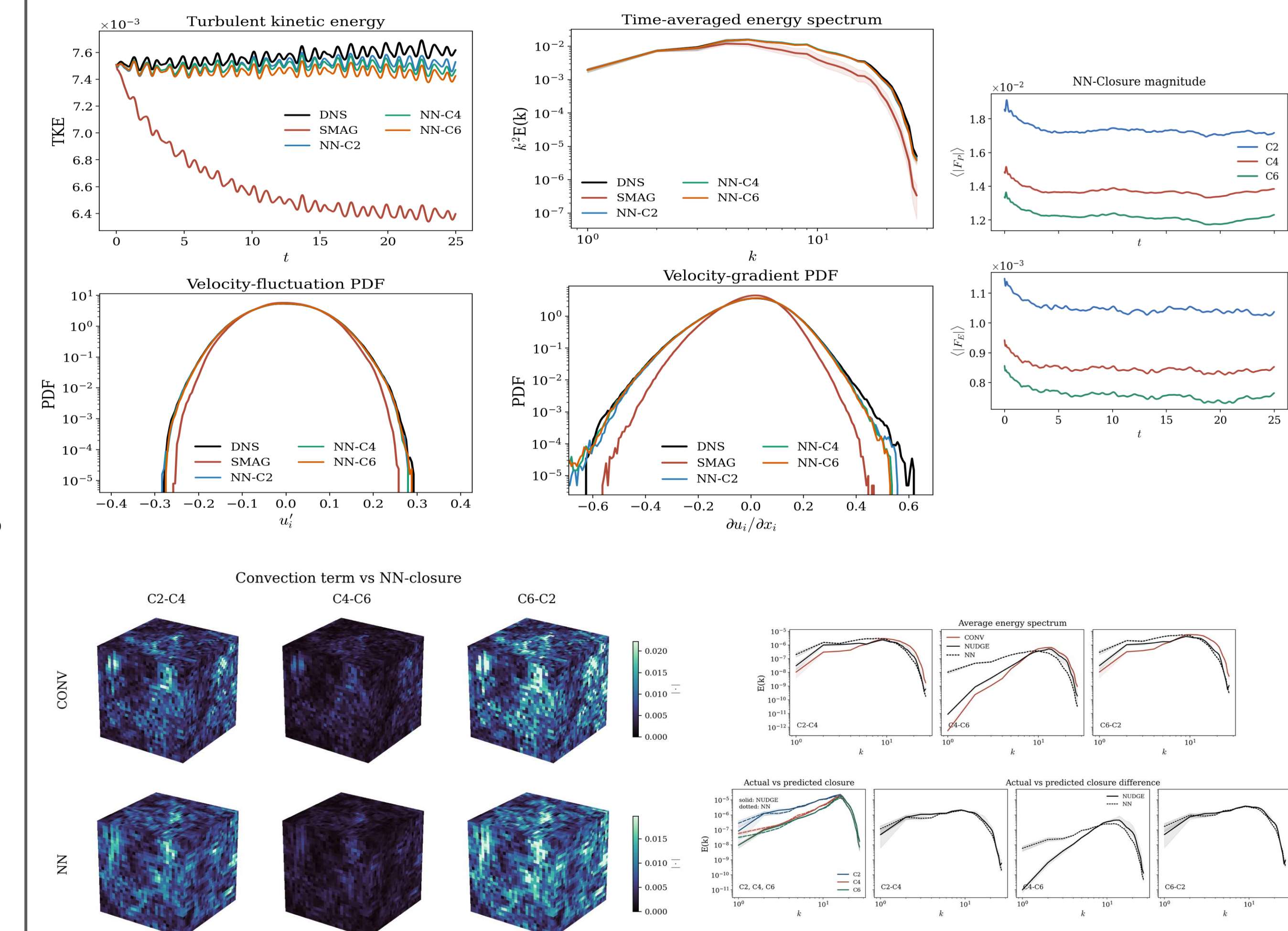


Results : 3D HIT



In 3D, we use compressible FVM solver, the forcing is done in its conservative form as:

$$F_{momentum} = \mu(\rho_{obs} u_{obs} - \rho v); F_{energy} = v \cdot F_{momentum}$$



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